

Overview of E1733A Calibration Software Operation

Contents

Introduction	2
Measurement Selection Screen	3
E1733A Program Setup	4
Basic Measurement Screens	5
System Setup	8
Measurement Setup	11
Machine Information Setup	13
ISO Information Setup	14
Environmental Compensation Setup	15
Data Analysis Setup	16
Graph Screen	17
Data Screen	18
Comp. Screen	19
Env. Screens	20
Timebase Screen	21
Flatness Screen	23

Overview of E1733A Calibration Software Operation

Introduction

The E1733A Calibration Software is Keysight's latest software for collecting and analyzing position data using the Keysight 5530 measurement system. Note that this system uses all the same optics as the 5529 system, so all the optical setups are the same as described in the manuals and help files for that system and the 10747* software. What follows is an overview of E1733A software and its operation. It should be considered a supplement to the other manuals.

Overview of E1733A Calibration Software Operation

Measurement Selection Screen

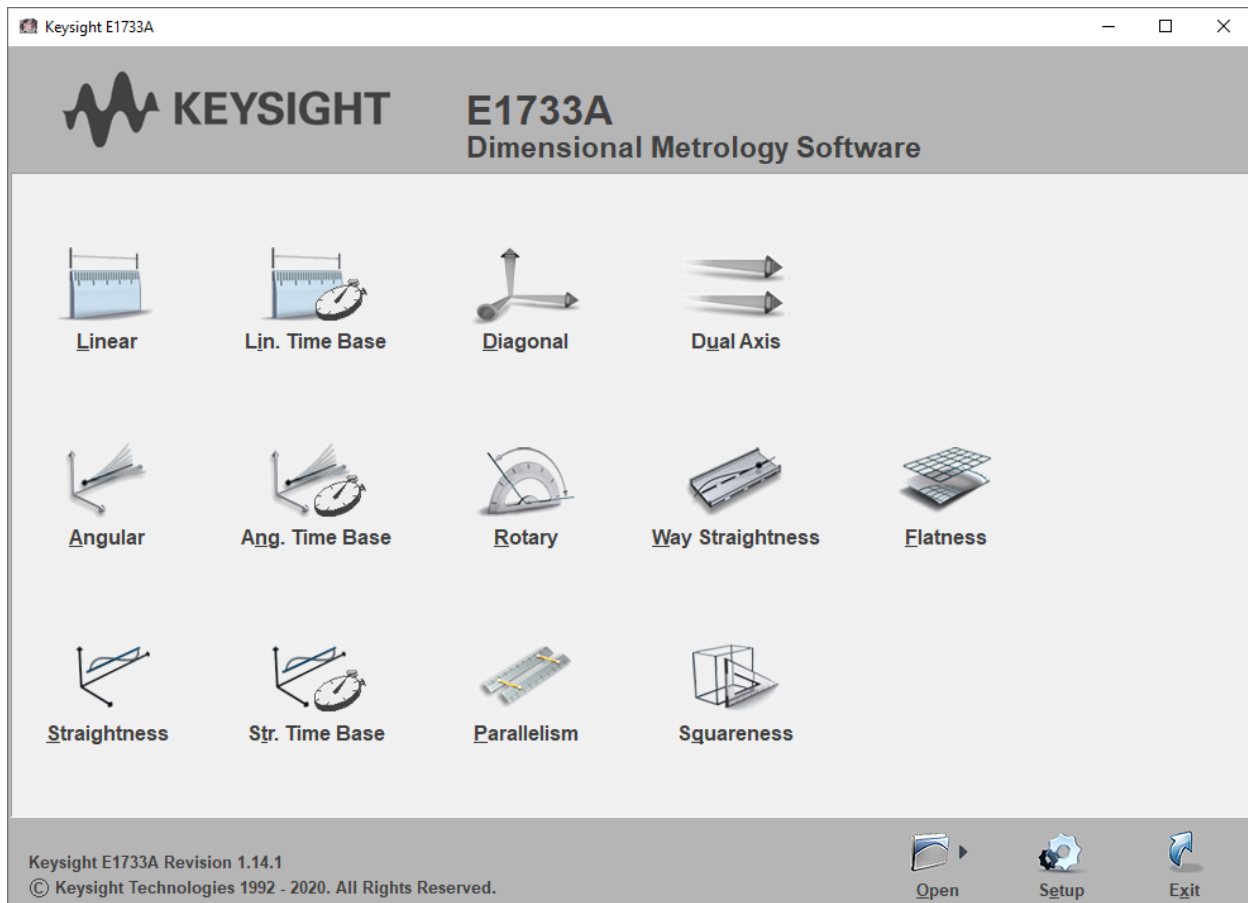
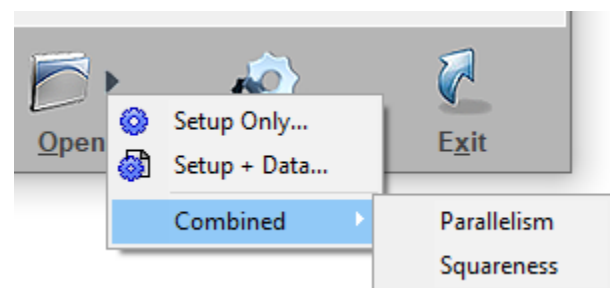


Figure 1. Measurement Selection Screen

The above figure shows the main screen that is displayed after starting the E1733A Calibration Software. Note the revision number in the bottom left corner. Please include this information with any requests for assistance or reports of problems encountered. This main screen allows quick selection of the:

- desired measurement (position mouse over the desired measurement selection and left click).
- language selection, default data directory, or method for pre-populating fields in new measurements (click on **Setup**).
- prior measurement data file(s) (click on **Open**). Note that the “Combined” options will sequentially request two straightness data files for combination into a single file.



Overview of E1733A Calibration Software Operation

E1733A Program Setup

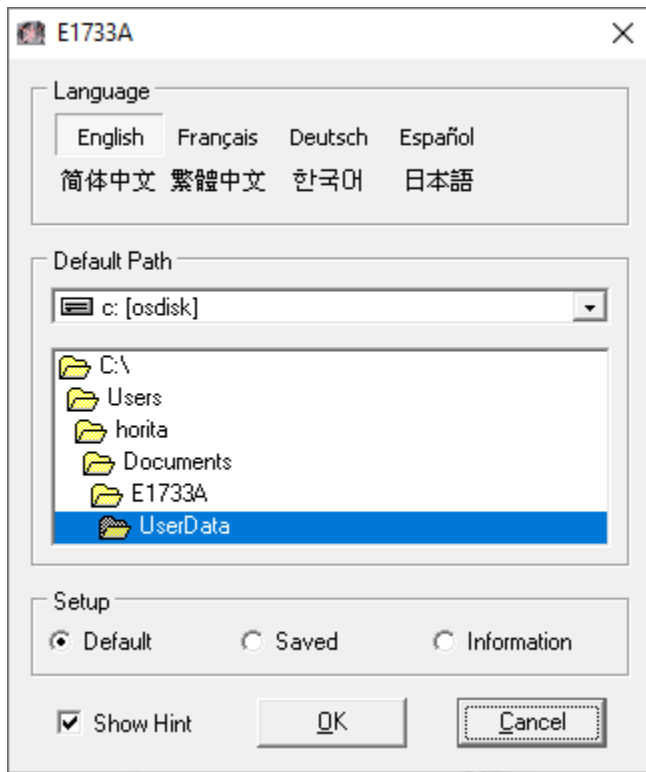


Figure 2. E1733A Program Setup dialog

The “Program Setup” dialog can be used to select an alternate language for all displays, along with selecting the default data file location for **Save** and **Open** operations. The lower “Setup” portion of this dialog specifies how some measurement setup fields are pre-populated when starting a new measurement:

- **Default:** “Measurement Setup” fields are pre-populated with default setup information for the selected measurement type. All other fields (“Machine Information” and “ISO Information”) are left blank.
- **Saved:** “Measurement Setup” information from the most recent measurement is pre-populated. All other fields (“Machine Information” and “ISO Information”) are left blank.
- **Information:** “Measurement Setup” fields are pre-populated with default setup information for the selected measurement type. The “Machine Information” and “ISO Information” fields are pre-populated with values from the most recent measurement.

Overview of E1733A Calibration Software Operation

Basic Measurement Screens

Figure 3 thru Figure 14 show the setup and measurement screens for Linear measurements. Other measurements selected from the main selection screen have similar sets of screens. Note that Timebase measurements and Flatness measurement have different appearance and menu.

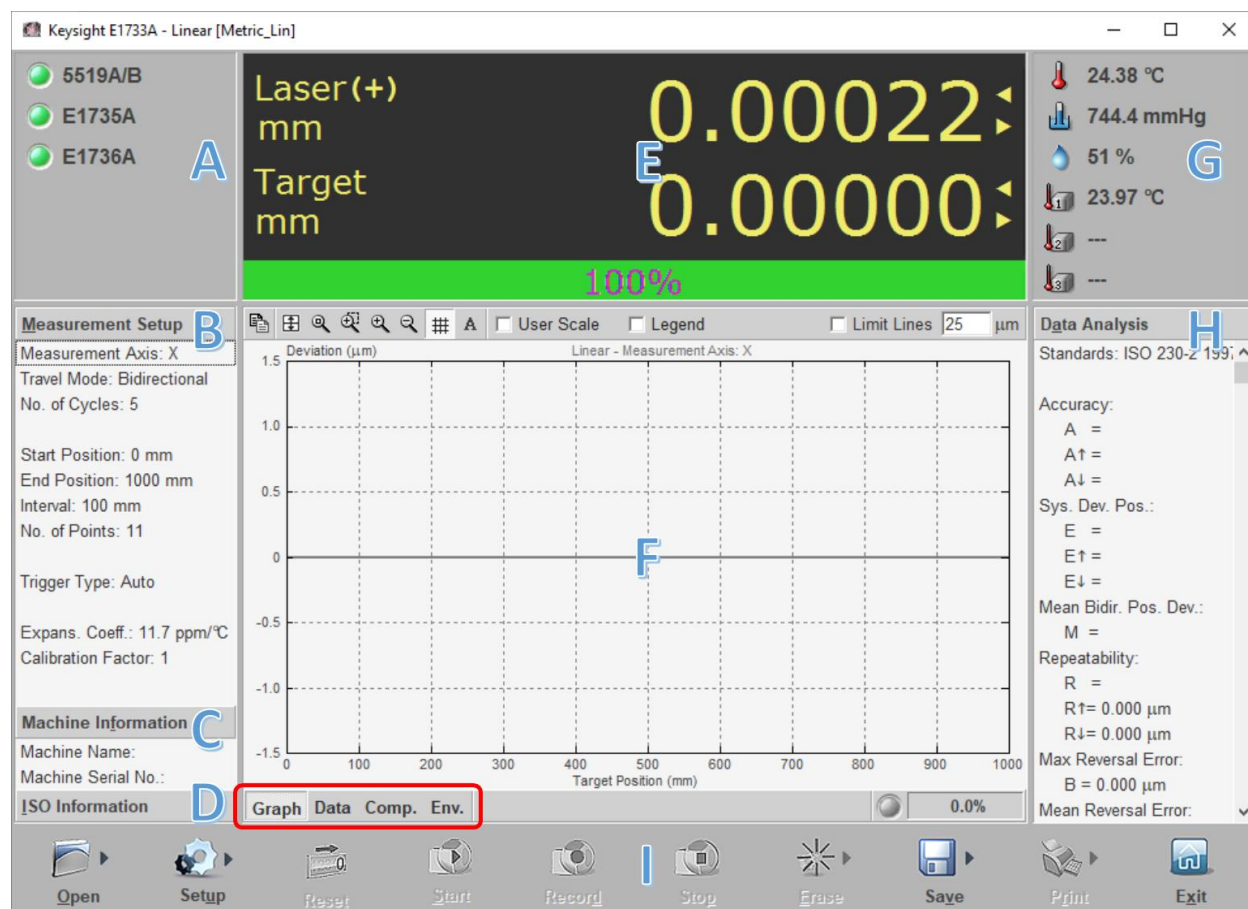


Figure 3. Linear Measurement Screen

Each section of the above figure operates as follows:

A. System:

This section presents status of each products. The indicators to the left of each product number will be green, yellow, or gray to reflect their status (found and OK, found but has an issue, or not found respectively).

Single clicking any item or double clicking anywhere in this area brings up the “System” dialog shown in Figure 4. See the section, [System Setup](#).

B. Measurement Setup:

This section presents measurement condition setup.

Single clicking this area’s title bar or double clicking anywhere in this area brings up the “Measurement Setup” dialog shown in Figure 5. See the section,

Overview of E1733A Calibration Software Operation

Measurement Setup.

C. Machine Information:

This section presents entries for machine name and serial numbers.

Single clicking this area's title bar or double clicking anywhere in this area brings up the "Machine Information" dialog shown in [Figure 6](#). See the section, [Machine Information Setup](#).

D. ISO Information:

Single clicking this title bar brings up the "ISO Information" dialog as shown in [Figure 7](#). See the section, [ISO Information Setup](#).

E. Position and Error Display:

The Position and Error Display area has multiple operations, but in general, if the item changes color when the mouse is moved over it, then clicking on the item will perform some function:

- a) Double clicking in this area makes the Position and Error Display occupy the entire window other than the bottom toolbar.
- b) Clicking the (+) sign changes the laser direction sense and will show as (-) after being clicked.
- c) Clicking "Laser" changes the display to "Error"
- d) Clicking "Target" changes the display to "Encoder" (only if the "Trigger Type" in the "Measurement Setup" dialog is set to "Encoder").
- e) Clicking the arrows to the right of the numbers increase and decrease the number of decimal places displayed for each value.

F. Graph and Data Screen:

The Measurement Graph and Data area has multiple operations as follows:

- a) Double clicking in this area makes the Measurement Graph and Data area occupy the entire window other than the bottom toolbar.
- b) Clicking the small speed-bar buttons at the top perform some copy and zoom operations and toggle the graph grid display and "Error Sign Convention" settings.
- c) Clicking the tabs at the bottom (boxed in red) select alternate ways (Graphical, Data Table, Compensation Table, Environmental Data – Statistics, Temp Graph, Pressure Graph, RH Graph, Material Temp Average Graph) to view the collected data (see [Figure 10](#) thru [Figure 14](#)).

Note that Timebase measurements and Flatness measurement have different appearance and menu. See the sections, [Timebase Screen](#) and [Flatness Screen](#).

G. Environmental Compensation:

This section presents environmental data, temp, humidity and pressure of atmosphere plus material temperatures.

A grayed-out icon and value indicate that that value is set manually on the Environmental screen instead of automatically by reading the sensor (indicated by a full color icon and a non-grayed-

Overview of E1733A Calibration Software Operation

out value).

Single clicking any item or double clicking anywhere in this area brings up the “Environmental Compensation” screen shown in [Figure 8](#). See the section, [Environmental Compensation Setup](#).

H. Data Analysis:

Single clicking this area’s title bar or double clicking anywhere in this area brings up the “Data Analysis” setup dialog shown in [Figure 9](#). See the section, [Data Analysis Setup](#).

I. Toolbar:

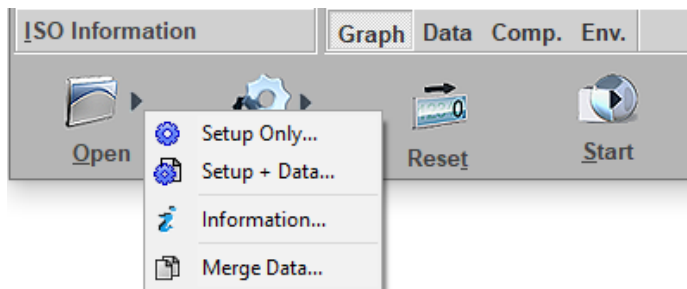
The toolbar at the bottom of the screen is designed to guide the user through the steps for making a measurement by following the buttons from left to right.

One can **Open** and **Save** files, **Print** reports, **Start** and **Stop** the measurement process, and **Erase** data by using the toolbar buttons at the bottom of the screen. (see below)



Note that when saving either the raw data or the compensation table data using the Save button, the file type drop down list allows one to select the sort order (run number or position number) for saving the data.

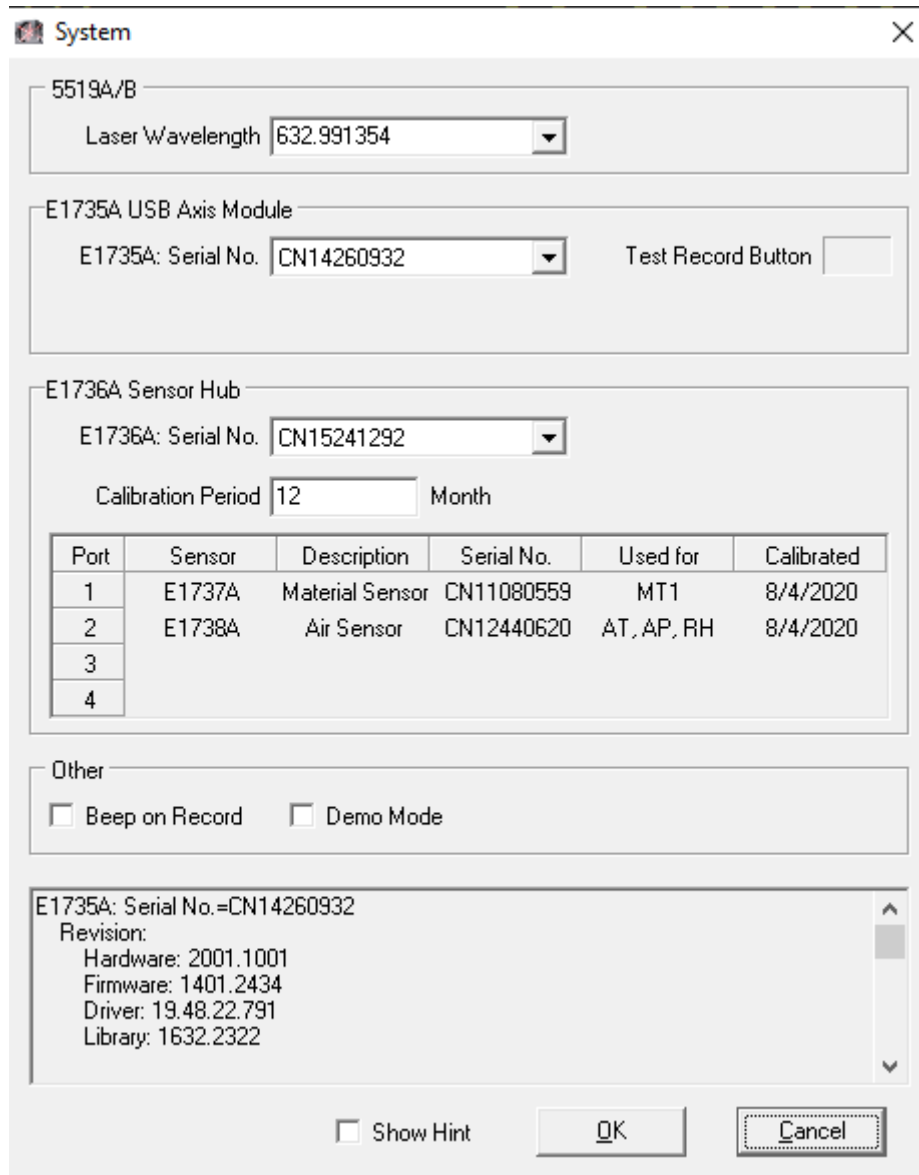
The “Merge Data ...” option in the popup menu from the Open button allows one to compare two linear/angular/straightness measurements having the same type and target lists.



Note that any Save operations after using the Merge selection only operate on data from the 1st file loaded.

Overview of E1733A Calibration Software Operation

System Setup



System [X]

5519A/B

Laser Wavelength: 632.991354

E1735A USB Axis Module

E1735A: Serial No.: CN14260932 Test Record Button

E1736A Sensor Hub

E1736A: Serial No.: CN15241292

Calibration Period: 12 Month

Port	Sensor	Description	Serial No.	Used for	Calibrated
1	E1737A	Material Sensor	CN11080559	MT1	8/4/2020
2	E1738A	Air Sensor	CN12440620	AT, AP, RH	8/4/2020
3					
4					

Other

☐ Beep on Record ☐ Demo Mode

E1735A: Serial No.=CN14260932

Revision:

Hardware: 2001.1001

Firmware: 1401.2434

Driver: 19.48.22.791

Library: 1632.2322

☐ Show Hint OK Cancel

Figure 4. System dialog

The “System” dialog allows one to view information about the hardware connected to the system and to change the laser wavelength being used by the software.

Overview of E1733A Calibration Software Operation

In addition, the dialog allows one to set the Calibration Period (month) for sensors. In case the sensor is over due to the period from the calibration last time, the software alert users.

E1736A Sensor Hub

E1736A: Serial No.

Calibration Period Month  Calibration Expired

Port	Sensor	Description	Serial No.	Used for	Calibrated
1	E1737A	Material Sensor	CN11080559	MT1	 3/30/2012
2	E1738A	Air Sensor	CN12440620	AT, AP, RH	 9/29/2014
3					
4					

Note that this dialog contains the “Beep on Record” check box, allowing it to be turned off or on during a measurement without interrupting the measurement.

2	E1738A	Air Sensor	CN12440620	AT, AP, RH	8/4/2020
3					
4					

Other

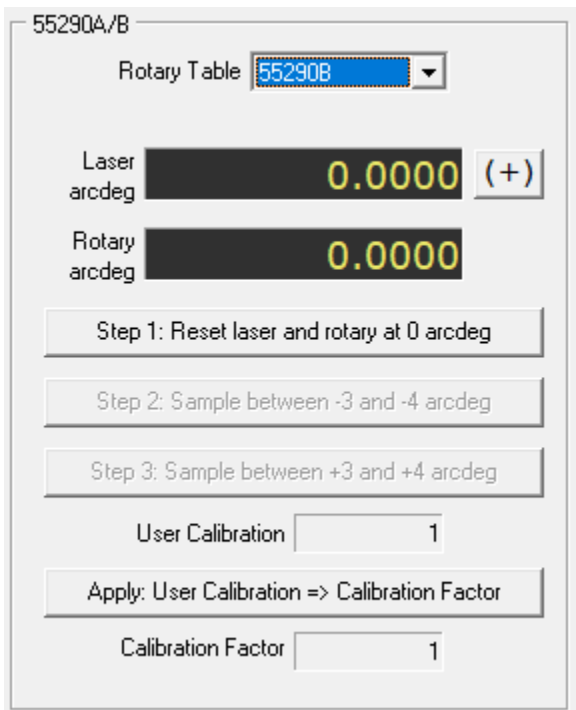
☐ Beep on Record ☐ Demo Mode

E1735A: Serial No.=CN14260932
Revision:

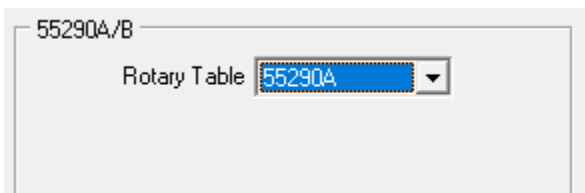
The “Demo Mode” checkbox allows one to experiment with the software without having any laser hardware connected to the computer. Most of the software is operational, however the encoder position is not simulated, and thus the Encoder Trigger should not be selected when using Demo mode.

Overview of E1733A Calibration Software Operation

For Rotary measurements, use the additional 55290A/B section to specify which rotary table is being used.



If the 55290B is selected, follow steps 1-3 to obtain an Angular Optics calibration factor. Click the “Apply: User Calibration => Calibration Factor” to accept this Calibration Factor for use with the rotary measurement.



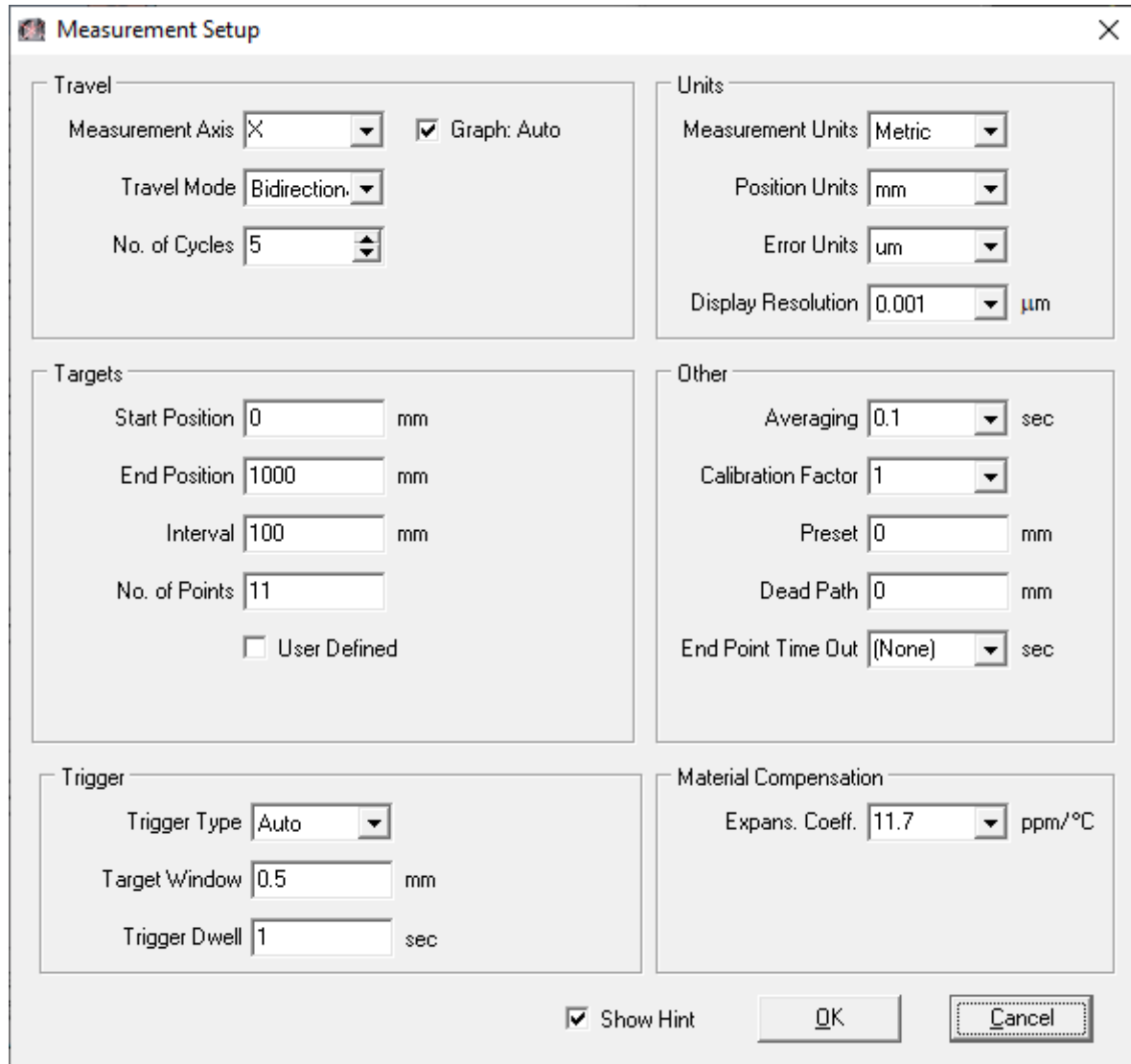
When using the 55290A table, one can manually perform the same calibration. Lock the base in place and rotate just the top portion of the indexing table to – 3 degrees as measured by the indexing table, noting the laser position as L1. Then move the top portion of the table to +3 degrees, noting the laser position as L2. Compute the Optical Calibration Factor:

$$OCF = \text{ABS}((L2-L1) / (E2-E1)), \quad E2-E1 = 6 \text{ for } 55290A \text{ at } +/- 3 \text{ degrees}$$

Then enter that factor on the Measurement Setup dialog.

Overview of E1733A Calibration Software Operation

Measurement Setup



The screenshot shows the "Measurement Setup" dialog box with the following settings:

- Travel:**
 - Measurement Axis: X
 - Graph: Auto (checked)
 - Travel Mode: Bidirection.
 - No. of Cycles: 5
- Units:**
 - Measurement Units: Metric
 - Position Units: mm
 - Error Units: μm
 - Display Resolution: 0.001 μm
- Targets:**
 - Start Position: 0 mm
 - End Position: 1000 mm
 - Interval: 100 mm
 - No. of Points: 11
 - User Defined (unchecked)
- Other:**
 - Averaging: 0.1 sec
 - Calibration Factor: 1
 - Preset: 0 mm
 - Dead Path: 0 mm
 - End Point Time Out: (None) sec
- Trigger:**
 - Trigger Type: Auto
 - Target Window: 0.5 mm
 - Trigger Dwell: 1 sec
- Material Compensation:**
 - Expans. Coeff.: 11.7 ppm/ $^{\circ}\text{C}$

At the bottom, there is a "Show Hint" checkbox (checked), and "OK" and "Cancel" buttons.

Figure 5. Measurement Setup dialog

The "Measurement Setup" dialog is used to specify how the measurement will be made. Note that although only X, Y, and Z are listed in the drop-down list for "Measurement Axis", any letter or designation can be typed into this field, and any changes made will be reflected in the title of the Graph in that section of the display.

Keysight recommends that the upper right "Measurement Units" selection be made prior to setting any of the "Targets" values.

Any time the "Targets" values are inconsistent with themselves, an "Inconsistent values" error message and recalculate buttons will be displayed to enable automatic adjustment of the desired value. Other setup error messages may also be displayed below the "User Defined" checkbox when certain encoder

Overview of E1733A Calibration Software Operation

setup conditions are not met. These errors must be fixed by manually adjusting the values referenced in the error message.

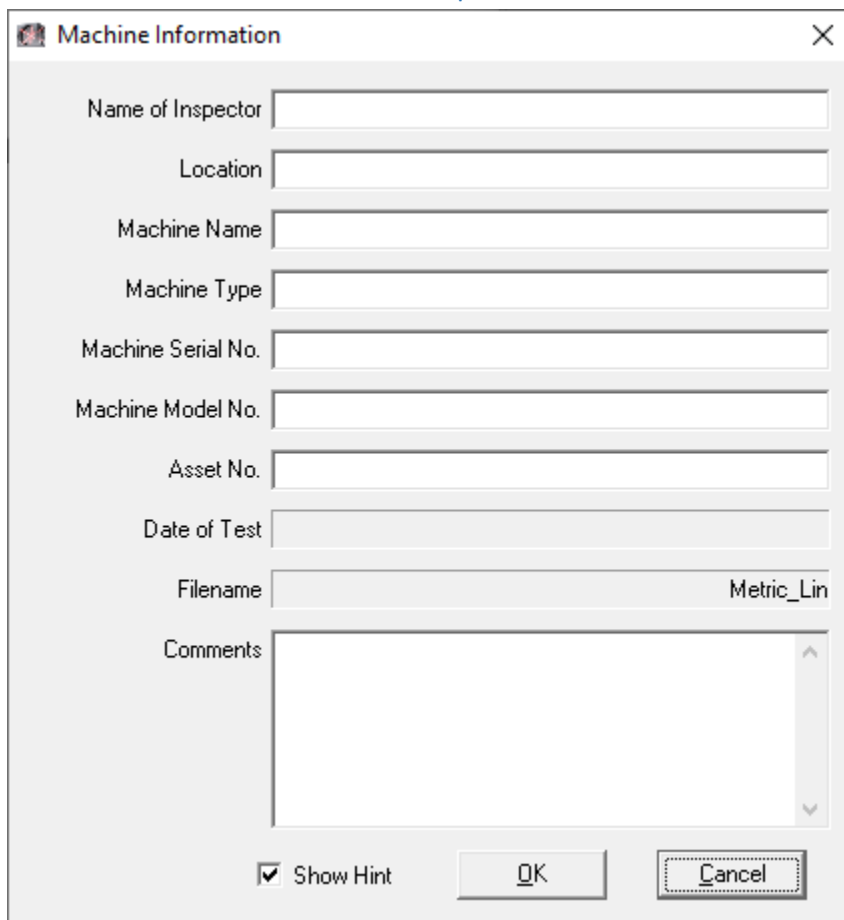
Checking the “User Defined” checkbox (which is only visible when the “Trigger Type” is not “Encoder”) effectively presses the “Interval” **Recalculate** button if it was displayed, and then displays the resulting target values, which can then be edited. An **Apply** button is also shown and should be pressed after editing the target values. Note that setting the “Trigger Type” to “Encoder” removes any “User Defined” targets and reverts to uniform targets as defined by the values in that section.

To enable adjustment of the target positions during a measurement, the “Trigger Type” must be set to “Manual” and the “Keyboard Position” checkbox must be checked. Then, when the laser position is within the target window, the Target Value will be highlighted in blue in the Position and Error Display section and can be edited. Pressing **Record** locks in both the target and laser measurement values.

When the “Trigger Type” is set to “Auto”, the “End Point Time Out” value is shown. It specifies how long to wait at an end point before auto triggering a 2nd time for the 1st point in the next run in the opposite direction. Note that “None” means it will not timeout, and thus the stage must move past the last point and then come back to it.

Overview of E1733A Calibration Software Operation

Machine Information Setup



The "Machine Information" dialog box is a standard Windows-style window with a title bar that says "Machine Information" and a close button (X). It contains several text input fields for the following information:

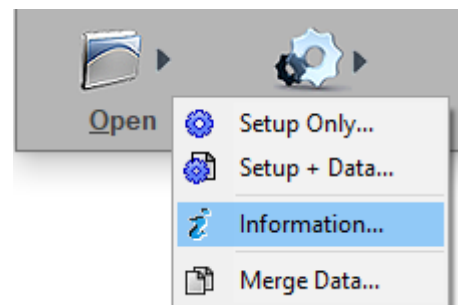
- Name of Inspector
- Location
- Machine Name
- Machine Type
- Machine Serial No.
- Machine Model No.
- Asset No.
- Date of Test
- Filename (with a "Metric_Lin" button next to it)
- Comments (a large text area with a scroll bar)

At the bottom of the dialog, there is a checkbox labeled "Show Hint" which is checked, and two buttons: "OK" and "Cancel".

Figure 6. Machine Info dialog

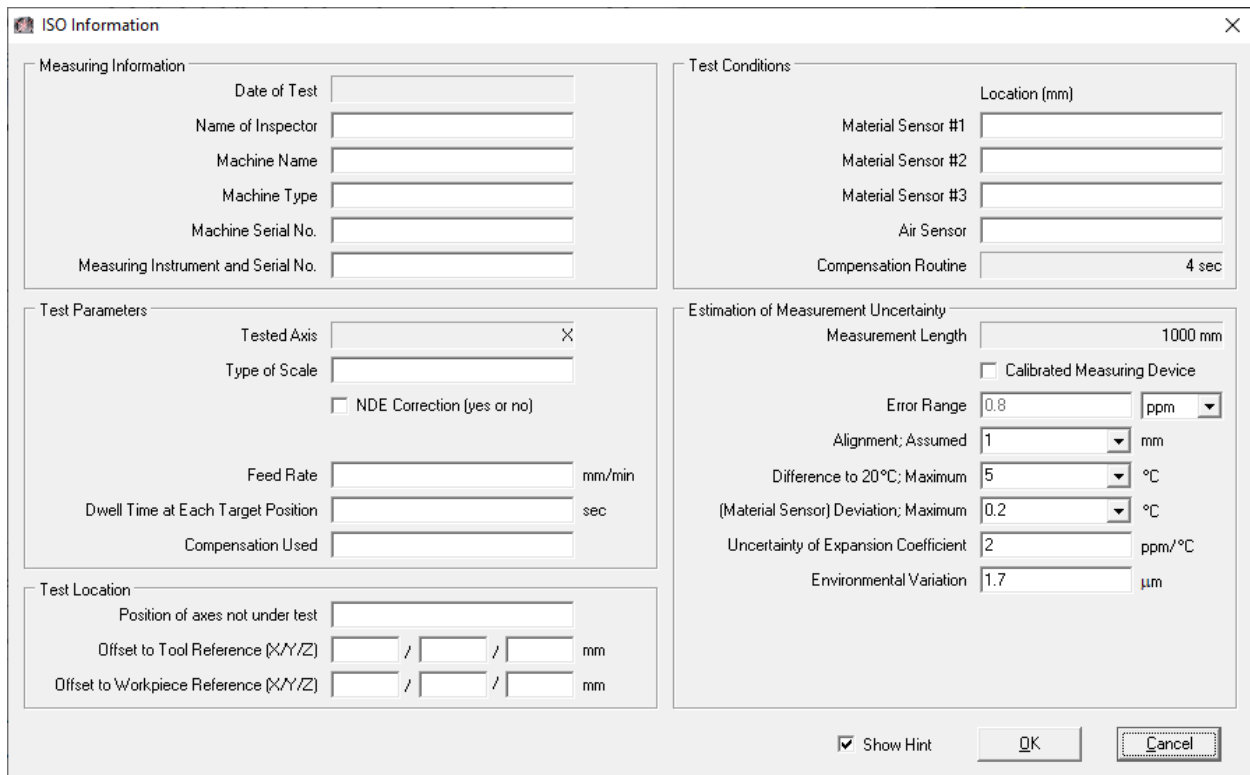
The "Machine Information" dialog allows one to enter any additional information as needed. The fields in this dialog can be pre-populated for new measurements in various ways. See [E1733A Program Setup](#) for additional details.

Additionally, these fields can be copied from another measurement file into an active measurement by performing an **Open** / Information operation from the [Basic Measurement Screen](#).



Overview of E1733A Calibration Software Operation

ISO Information Setup



ISO Information

Measuring Information

Date of Test

Name of Inspector

Machine Name

Machine Type

Machine Serial No.

Measuring Instrument and Serial No.

Test Conditions

Location (mm)

Material Sensor #1

Material Sensor #2

Material Sensor #3

Air Sensor

Compensation Routine 4 sec

Test Parameters

Tested Axis X

Type of Scale

☐ NDE Correction (yes or no)

Feed Rate mm/min

Dwell Time at Each Target Position sec

Compensation Used

Test Location

Position of axes not under test

Offset to Tool Reference (X/Y/Z) / / mm

Offset to Workpiece Reference (X/Y/Z) / / mm

Estimation of Measurement Uncertainty

Measurement Length 1000 mm

☐ Calibrated Measuring Device

Error Range 0.8 ppm

Alignment; Assumed 1 mm

Difference to 20°C; Maximum 5 °C

(Material Sensor) Deviation; Maximum 0.2 °C

Uncertainty of Expansion Coefficient 2 ppm/°C

Environmental Variation 1.7 µm

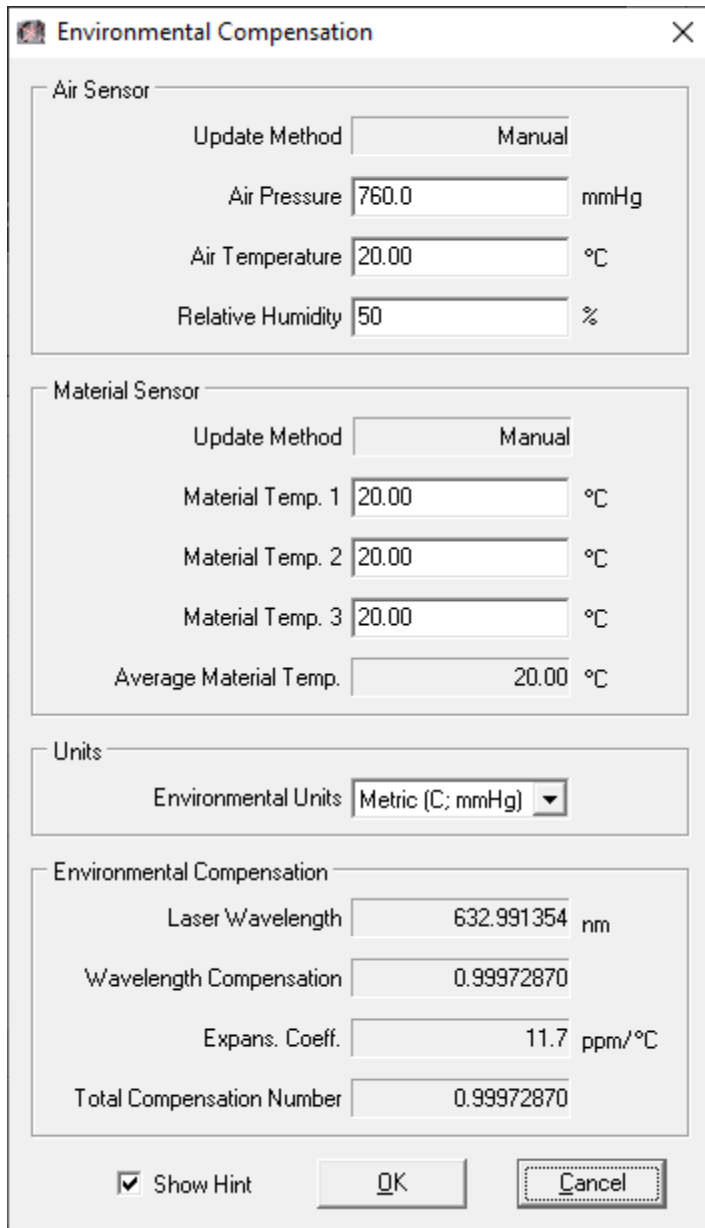
☒ Show Hint

Figure 7. Iso Info dialog

The “ISO Information” dialog allows one to see and adjust the information used for printing the ISO reports. The fields in this dialog can be pre-populated for new measurements in various ways. See [E1733A Program Setup](#) for additional details.

Overview of E1733A Calibration Software Operation

Environmental Compensation Setup



Environmental Compensation

Air Sensor

Update Method:

Air Pressure: mmHg

Air Temperature: °C

Relative Humidity: %

Material Sensor

Update Method:

Material Temp. 1: °C

Material Temp. 2: °C

Material Temp. 3: °C

Average Material Temp.: °C

Units

Environmental Units:

Environmental Compensation

Laser Wavelength: nm

Wavelength Compensation:

Expans. Coeff.: ppm/°C

Total Compensation Number:

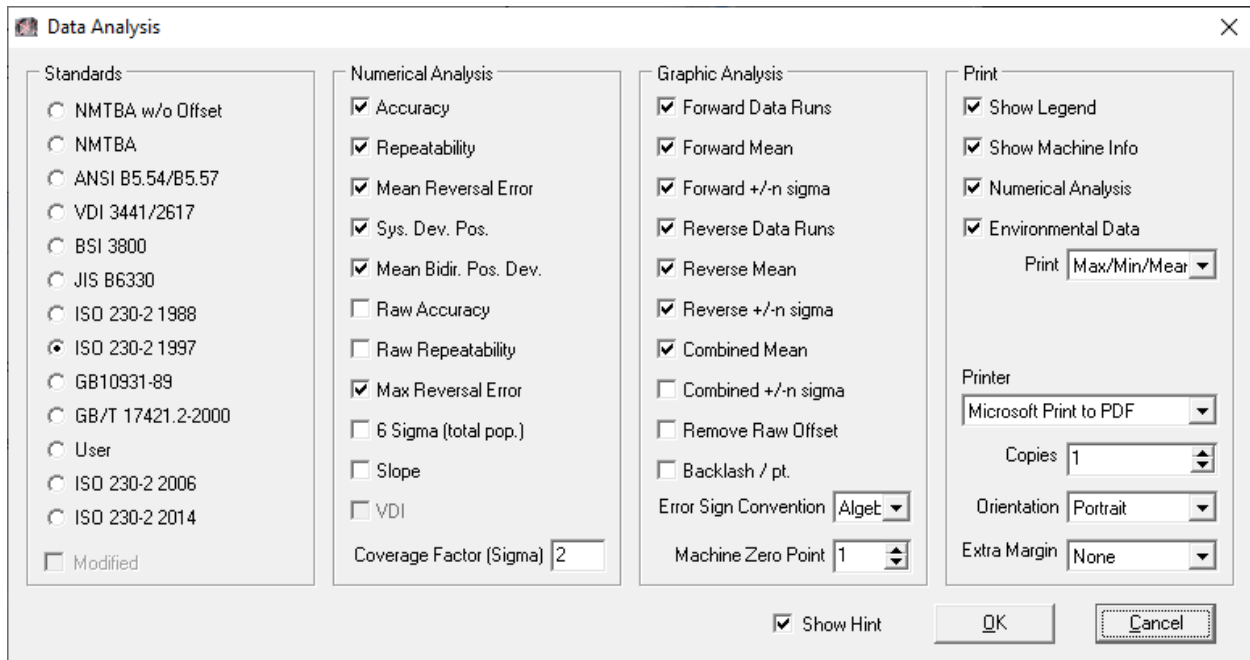
☒ Show Hint

Figure 8. Environmental Compensation Setup dialog

The “Environmental Compensation” dialog allows one to see (and manually adjust if desired) the current values being used to compensate the measurement for changes to the air density and material temperature.

Overview of E1733A Calibration Software Operation

Data Analysis Setup



Data Analysis

Standards

- ☐ NMTBA w/o Offset
- ☐ NMTBA
- ☐ ANSI B5.54/B5.57
- ☐ VDI 3441/2617
- ☐ BSI 3800
- ☐ JIS B6330
- ☐ ISO 230-2 1988
- ☒ ISO 230-2 1997
- ☐ GB10931-89
- ☐ GB/T 17421.2-2000
- ☐ User
- ☐ ISO 230-2 2006
- ☐ ISO 230-2 2014
- ☐ Modified

Numerical Analysis

- ☒ Accuracy
- ☒ Repeatability
- ☒ Mean Reversal Error
- ☒ Sys. Dev. Pos.
- ☒ Mean Bidir. Pos. Dev.
- ☐ Raw Accuracy
- ☐ Raw Repeatability
- ☒ Max Reversal Error
- ☐ 6 Sigma (total pop.)
- ☐ Slope
- ☐ VDI
- Coverage Factor (Sigma)

Graphic Analysis

- ☒ Forward Data Runs
- ☒ Forward Mean
- ☒ Forward +/-n sigma
- ☒ Reverse Data Runs
- ☒ Reverse Mean
- ☒ Reverse +/-n sigma
- ☒ Combined Mean
- ☐ Combined +/-n sigma
- ☐ Remove Raw Offset
- ☐ Backlash / pt.
- Error Sign Convention
- Machine Zero Point

Print

- ☒ Show Legend
- ☒ Show Machine Info
- ☒ Numerical Analysis
- ☒ Environmental Data
- Print
- Printer
- Copies
- Orientation
- Extra Margin

☒ Show Hint

Figure 9. Data Analysis dialog

The “Data Analysis” dialog allows one to specify the standard to use for the analysis, and to optionally customize the selected standard by modifying the resulting settings in the middle and right panels of the dialog.

One can also specify which printer to use by default for printing reports, along with orientation and adding additional margins to facilitate keeping printed reports in a binder.

Overview of E1733A Calibration Software Operation

Graph Screen

Graph shows the Position data for all measurements except Timebase and Flatness measurements.

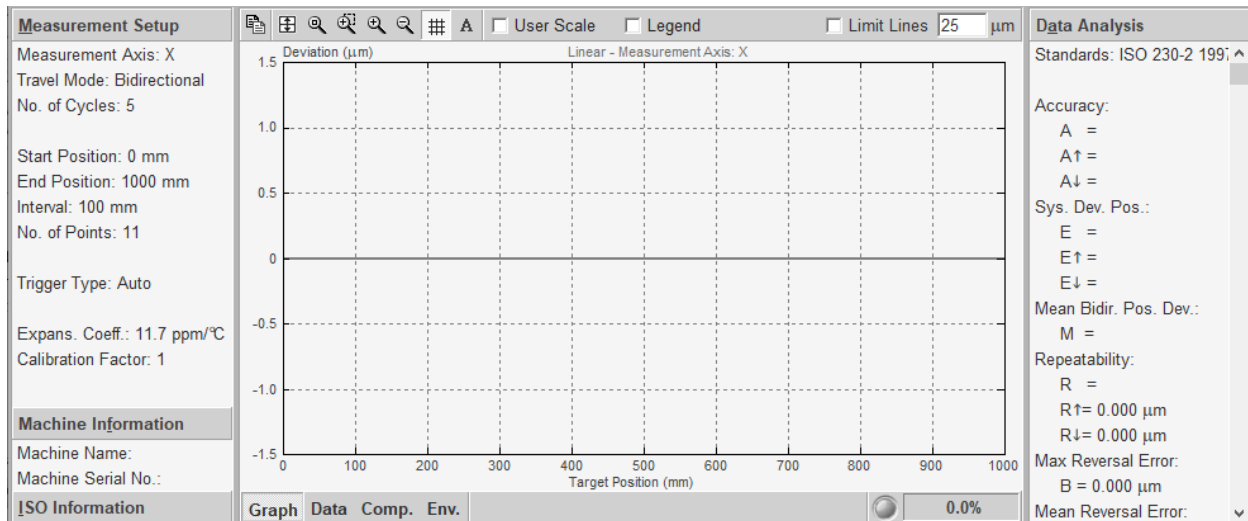


Figure 10. Linear Measurement

The speed buttons at the top allow one to:



- : Copy the graph to the clipboard.
- : Display the graph with equal '+' and '-' ranges.
- : Display the full graph with optimum zoom.
- : Zoom into a specific area that one selects with the mouse.
- : Zoom in and out.
- : Turn the grid lines on and off.
- : Change the "Error Sign Convention" selection.

One can also set the scaling directly, display or hide a draggable legend box, and show +/- limit lines at the specified location. Furthermore, one can click and drag the graph as desired using the left or right mouse buttons.

Overview of E1733A Calibration Software Operation

Data Screen

Measurement Setup		1 +X A ☐ Limit Lines 25 μm								Data Analysis																																																																														
Measurement Axis: X Travel Mode: Bidirectional No. of Cycles: 5 Start Position: 0 mm End Position: 1000 mm Interval: 100 mm No. of Points: 11 Trigger Type: Auto Expans. Coeff.: 11.7 ppm/ $^{\circ}\text{C}$ Calibration Factor: 1		<table border="1"> <thead> <tr> <th></th> <th>μm</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tr> <tr> <th></th> <th></th> <th></th> <th></th> <th></th> <th></th> </tr> </thead> <tbody> <tr><td>1</td><td>0</td><td></td><td></td><td></td><td></td></tr> <tr><td>2</td><td>100</td><td></td><td></td><td></td><td></td></tr> <tr><td>3</td><td>200</td><td></td><td></td><td></td><td></td></tr> <tr><td>4</td><td>300</td><td></td><td></td><td></td><td></td></tr> <tr><td>5</td><td>400</td><td></td><td></td><td></td><td></td></tr> <tr><td>6</td><td>500</td><td></td><td></td><td></td><td></td></tr> <tr><td>7</td><td>600</td><td></td><td></td><td></td><td></td></tr> <tr><td>8</td><td>700</td><td></td><td></td><td></td><td></td></tr> <tr><td>9</td><td>800</td><td></td><td></td><td></td><td></td></tr> <tr><td>10</td><td>900</td><td></td><td></td><td></td><td></td></tr> <tr><td>11</td><td>1000</td><td></td><td></td><td></td><td></td></tr> </tbody> </table>									μm	1	2	3	4							1	0					2	100					3	200					4	300					5	400					6	500					7	600					8	700					9	800					10	900					11	1000					Standards: ISO 230-2 199: ^ Accuracy: A = A $\uparrow$ = A $\downarrow$ = Sys. Dev. Pos.: E = E $\uparrow$ = E $\downarrow$ = Mean Bidir. Pos. Dev.: M = Repeatability: R = R $\uparrow$ = 0.000 μm R $\downarrow$ = 0.000 μm Max Reversal Error: B = 0.000 μm Mean Reversal Error: v
	μm	1	2	3	4																																																																																			
1	0																																																																																							
2	100																																																																																							
3	200																																																																																							
4	300																																																																																							
5	400																																																																																							
6	500																																																																																							
7	600																																																																																							
8	700																																																																																							
9	800																																																																																							
10	900																																																																																							
11	1000																																																																																							
Machine Information Machine Name: Machine Serial No.:		< 0.0%																																																																																						
ISO Information																																																																																								

Figure 11. Data Screen

The “Data” screen shows all the collected data in tabular form. One can click on a cell to select it and then edit the value in that cell if needed (only the 1st edit requires confirmation). The down and up arrows near the top correspond with reverse and forward travel directions respectively.

Additionally, the speed buttons at the top allow one to:

	1			+X	A	☐ Limit Lines	25 μm
--	---	--	--	----	---	--------------------------------------	------------------

- : Copy the entire table to the clipboard (tab separated columns with ‘-’ and ‘+’ symbols used in the heading to indicate reverse and forward travel directions).
- 1: Set the selected cell to the current column in the specified row.
- : Switch between runs in rows or columns. Note that run numbers are indicated with the symbol while point numbers use the symbol.
- : Change the sign of all the error values.
- : Change the “Error Sign Convention” selection.

Overview of E1733A Calibration Software Operation

Comp. Screen

Measurement Setup		Download		Data Analysis	
Measurement Axis: X	#	Target Position	Combined	Machine Zero Point	1
Travel Mode: Bidirectional	1	0		Error Sign Convention	Algebraic
No. of Cycles: 5	2	100		Compensation Values	Absolute
Start Position: 0 mm	3	200			Correction
End Position: 1000 mm	4	300			Combined
Interval: 100 mm	5	400		Measurement Units	Metric
No. of Points: 11	6	500		Machine Units	1 um
Trigger Type: Auto	7	600		Target Position	Target List
Expans. Coeff.: 11.7 ppm/°C	8	700			
Calibration Factor: 1	9	800			
	10	900			
	11	1000			
Machine Information					
Machine Name:					
Machine Serial No.:					
ISO Information					
Graph Data Comp. Env.				0.0%	

Standards: ISO 230-2 199; ^

Accuracy:

A =

A↑ =

A↓ =

Sys. Dev. Pos.: E =

E↑ =

E↓ =

Mean Bidir. Pos. Dev.: M =

Repeatability: R =

R↑ = 0.000 μm

R↓ = 0.000 μm

Max Reversal Error: B = 0.000 μm

Mean Reversal Error: v

Figure 12. Comp. Screen

The “Comp.” screen displays the comp table and allows one to adjust all the setup terms used to calculate it from the measurement data. The speed buttons at the top left allow one to copy the table to the clipboard and to enlarge and reduce the font size as desired. Those at the top right allow one to specify and launch another program on the pc which can be used to download the compensation table to the machine being calibrated.

Overview of E1733A Calibration Software Operation

Env. Screens

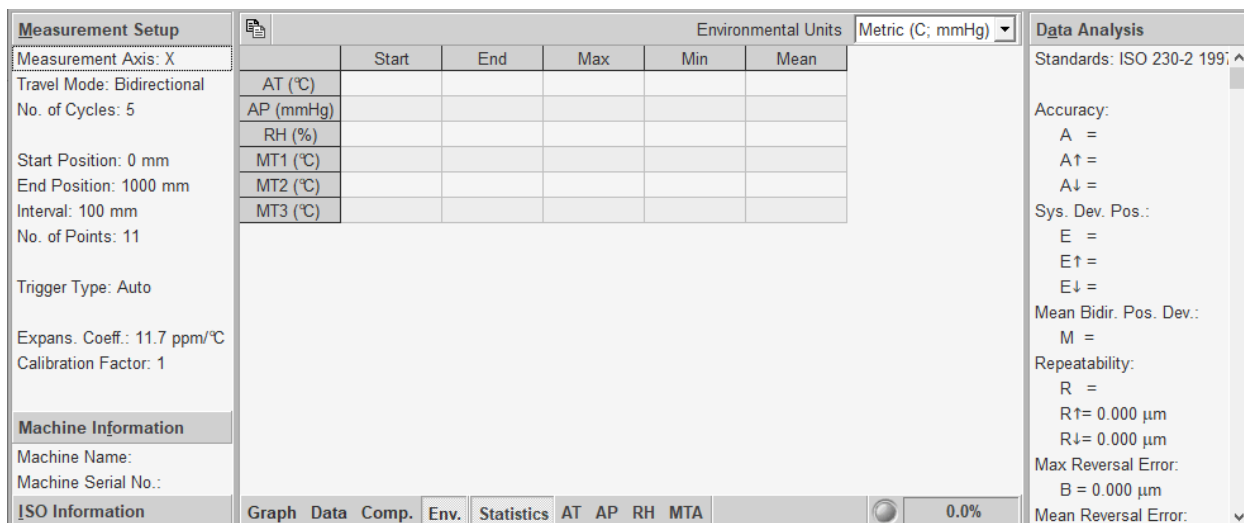


Figure 13. Env. Statistics Screen

There are multiple Environmental screens. One shows a summary of the data (Statistics) while the others show a time graph of the individual sensor readings (AT, AP, RH, and MTA). The speed buttons allow one to copy the statistics table or sensor graph to the clipboard and to zoom the graphs and display and hide the grid lines (see speed button descriptions under [Graph Screen](#)). Additionally, one can right or left click on the graph and drag the image as desired to see other areas once zoomed in.

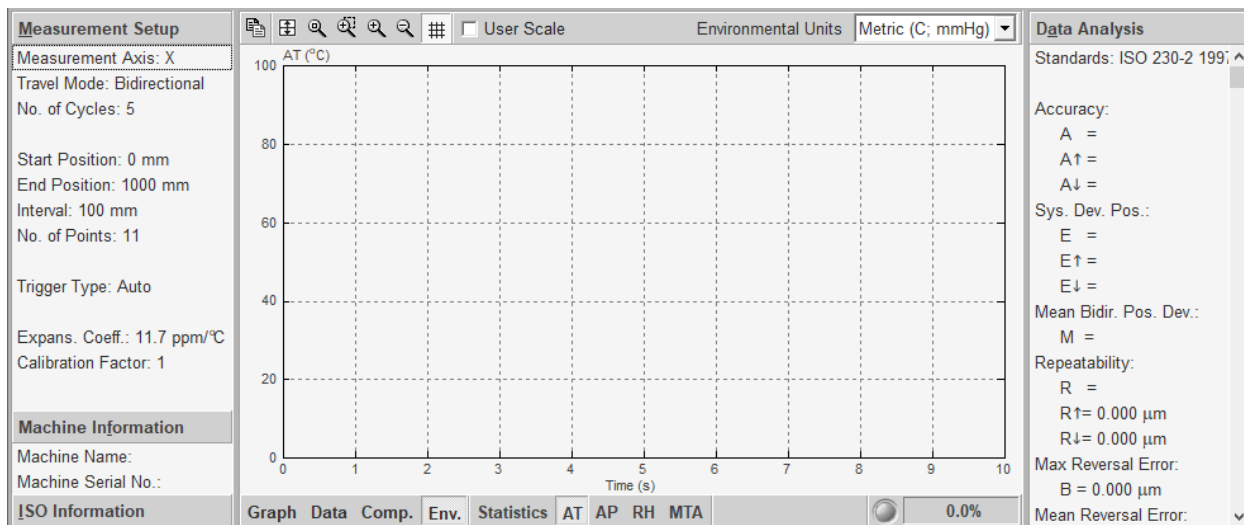


Figure 14. Env. Graph Screen

Overview of E1733A Calibration Software Operation

Timebase Screen

Timebase screen shows the Position data, Velocity and Acceleration, or FFT data.

Position only

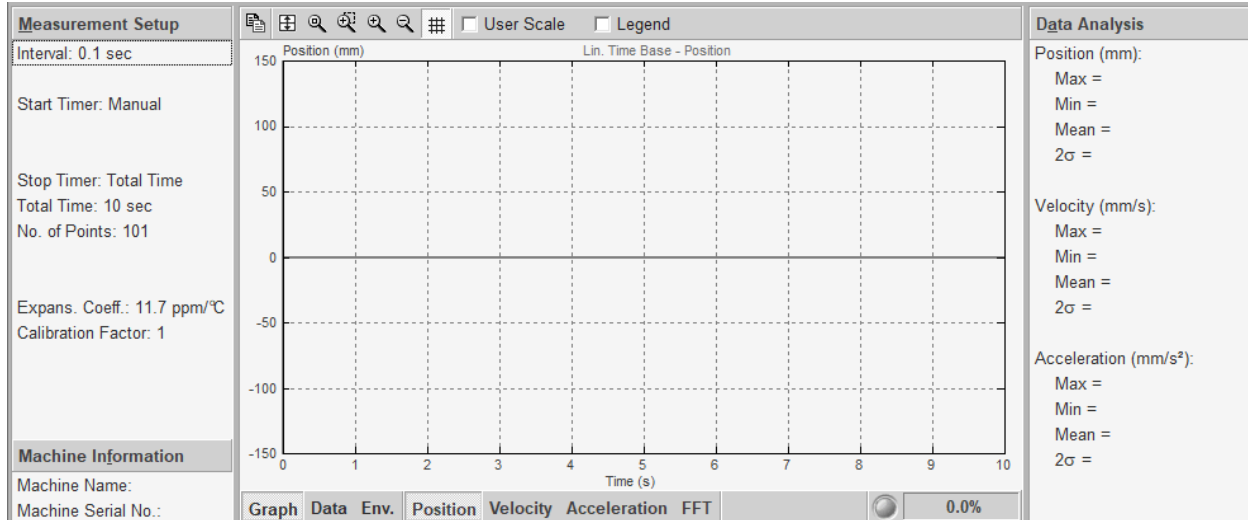


Figure 15. Timebase measurement

Multiple measurements

Timebase screen allow one to display multiple measurements, Position, Velocity and Acceleration in one screen except FFT. See [Figure 16](#).

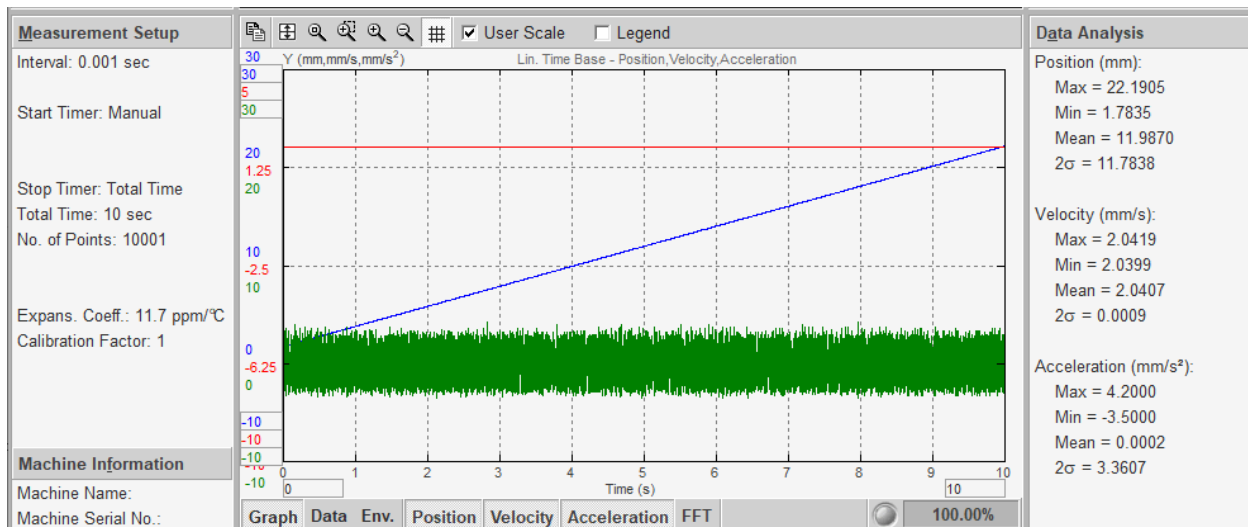


Figure 16. Multiple measurements, Position, Velocity and Acceleration

Overview of E1733A Calibration Software Operation

FFT mode

Timebase screen allow one to display FFT. See [Figure 17](#).

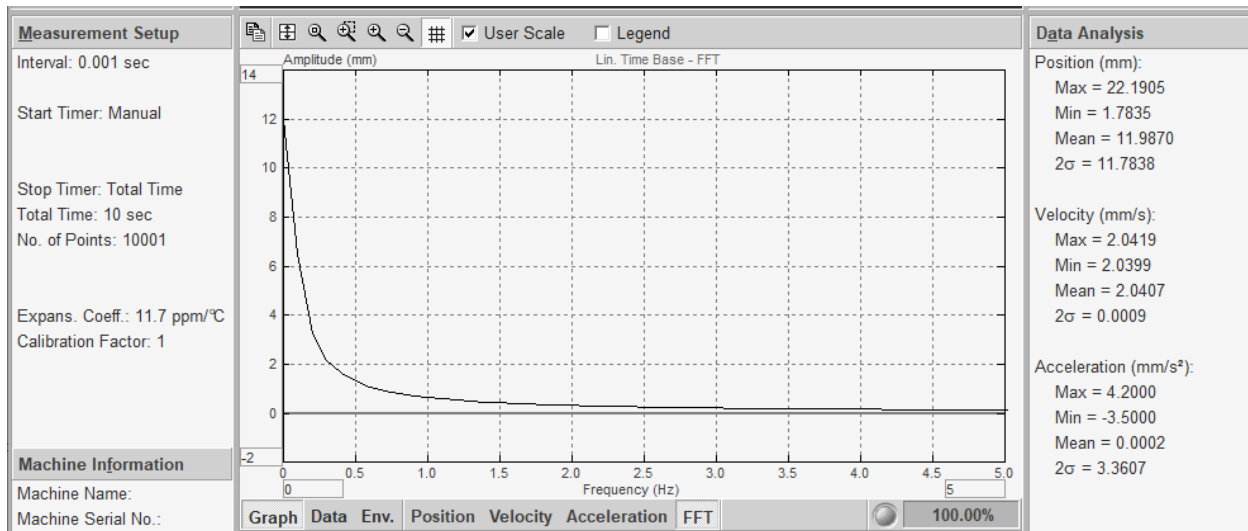


Figure 17. Timebase measurements, FFT mode

Overview of E1733A Calibration Software Operation

Flatness Screen

The default Flatness screen shows a 3-D drawing of the flatness data. One can right or left click and drag around the drawing to rotate and tilt the 3-D image.

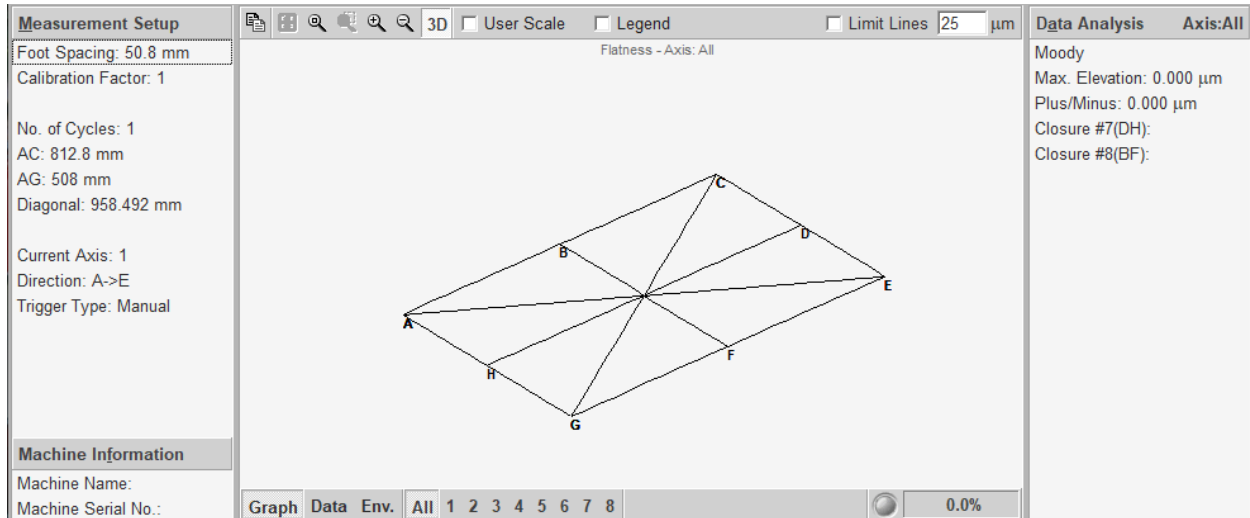


Figure 18. Flatness measurement

One can also use the  speed button at the top to replace the 3-D drawing with one having text showing the elevation at all the measurement points along each measurement line.

The bottom buttons allow showing the data for each line on a 2-D graph, or All the data on one of the drawings mentioned above, as well as seeing the data in table form, or the environmental data as shown in [Env. Screens](#).